FACTORS INFLUENCING STUDENTS' CHOICE OF STEM MAJOR IN NEPAL

A PROJECT WORK SUBMITTED TO THE

DEPARTMENT OF STATISTICS

TRI-CHANDRA MULTIPLE CAMPUS

INSTITUTE OF SCIENCE AND TECHNOLOGY

TRIBHUVAN UNIVERSITY

NEPAL

FOR THE AWARD OF

BACHELOR OF SCIENCE (B.Sc.) IN STATISTICS

BY

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AUGUST, 2024

RECOMMENDATION

This is to recommend that **CHIRAG DHAKAL SAHAJ,** (500370271, 5-2-37-153-2019), has carried out project work entitled "**FACTORS INFLUENCING STUDENTS' CHOICE OF STEM MAJOR IN NEPAL"** for the requirement to project work in Bachelor of Science (B.Sc.) degree in Statistics under my supervision in the Department of Statistics, Trichandra-Multiple Campus, Institution of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

To my/our knowledge, this work has not been submitted for any other degree.

He has fulfilled all the requirement laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the submission of the project work for the partial fulfillment of Bachelor of Science (B.Sc.) degree.

Nanda Kumar Tharu

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28 AUGUST, 2024

DECLARATION

This project work entitled "**FACTORS INFLUENCING STUDENTS' CHOICE OF STEM MAJOR IN NEPAL**" is being submitted to the Department of Statistics, Tri-Chandra Multiple Campus, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the partial fulfillment of the requirement to the project work in Bachelor of Science (B.Sc.) degree in Statistics. This project work is carried out by me under the supervision of Nanda Kumar Tharu in the Department of Statistics, Tri-Chandra Multiple Campus, Institute of Technology (IoST), Tribhuvan University (T.U.), Nepal.

This work is original and has not been submitted earlier in part or full in this or any other form to any university or institute, here or elsewhere, for the award of any degree.

Signature

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5-2-37-153-2019

28 AUGUST, 2024

LETTER OF FORWARD

Date: 28/08/2024

On the recommendation of **NANDA KUMAR THARU**, this project work is submitted by **CHIRAG DHAKAL SAHAJ**, 500370271**,** 5-2-37-153-2019, entitled "**FACTORS INFLUENCING STUDENTS' CHOICE OF STEM MAJOR IN NEPAL**" is forwarded by the Department of Statistics, Tri-Chandra Multiple Campus, for the approval to the Evaluation Committee, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

He has fulfilled all the requirements laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the project work.

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BOARD OF EXAMINATION AND

CERTIFICATE OF APPROVAL

This project work (PRO-406) entitled "**FACTORS INFLUENCING STUDENTS' CHOICE OF STEM MAJOR IN NEPAL**" by Chirag Dhakal Sahaj (500370271, 5-2-37-153-2019) under the supervision of Nanda Kumar Tharu in the Department of Statistics, Tri-Chandra Multiple Campus, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), is hereby submitted for the partial fulfillment of the Bachelor of Science (B.Sc.) degree in Statistics. This report has been accepted and forwarded to the Controller of Examination, Institute of Science and Technology, Tribhuvan University, Nepal for the legal procedure.

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Chirag Dhakal Sahaj

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ABSTRACT

The STEM acronym has been adjusted to include medicine because mathematics is considered to be a part of the Science wing in STEM in the context of Nepal. Nepal has recently experienced a rise in number of students pursuing STEM degrees. This study aims to understand the context of socio-economic, academic and personal variables in students' decision regarding the STEM major they pursue. Around 55 students were surveyed from different STEM colleges of Nepal, and those observations were used as prior distribution to generate 10000 observations using Markov Chain Monte Carlo (MCMC). The data analysis was done using Chi-square tests for socio-demographic differences and ANOVA for income and academic differences. Similarly, multinomial logistic regression model was used to explore association between variables and degree choice. The result indicates that academic factors like GPA and family related factors play a significant role in the selection of STEM major in Nepal. This study concludes that academic and personal factors play more role into students' decision regarding their choice of major than economic factors.

Keywords: Family influence, Markov Chain Monte Carlo (MCMC), Multinomial logistic regression, STEM Education

LIST OF ACRONYMS AND ABBREVIATIONS

ANOVA:	Analysis of Variance
CDC:	Curriculum Development Centre
GoN:	Government of Nepal
GPA:	Grade Point Average
HMC:	Hamiltonian Monte Carlo
MCMC:	Markov Chain Monte Carlo
NOC:	No Objection Certificates
NRB:	Nepal Rastra Bank
NSO:	National Statistics Office
NUTS:	No-U-Turn Sampler
PSRF:	Potential Scale Reduction Factors
SEE:	Secondary Education Examination
SCCT:	Social Cognitive Career Theory
STEM:	Science, Technology, Engineering and Medicine
UGC:	University Grants Commission

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

The STEM acronym has been modified in the Nepalese context to fit Nepal's social and educational sphere better. Instead of being viewed as a different field of study, mathematics is included under the wing of science in this research. Hence, the acronym is changed to include Medicine rather than Mathematics to appropriately reflect the social and educational context of STEM in Nepal. This change guarantees a more realistic portrayal of the subjects emphasized in Nepalese society discourse and the educational system. STEM is an acronym for Science, Technology, Engineering and Medicine.

The number of students majoring in STEM has surged in recent years in Asia, with China and India being the front runners producing the greatest number of STEM graduates (World Bank, 2022). Nepal was a late bloomer when it came to transitioning into the world of science and technology. Nepal established its first high education institute, Tri-Chandra College, in 1918. The college introduced the intermediate level, marking the inception of formal science education in Nepal. Later, after the democratization of the country in 1951, Tribhuvan University was established in 1959, starting university-level programs in natural sciences. (UNESCO, 2006).

Historically, economic growth has always been associated with capital and labor. However, after the advent of the Industrial Revolution, technology replaced the number of labourers and increased production efficiency. With this, technology usage requires a skilled workforce, and even further advancements in technology require innovation. Education derives a skilled workforce and breeds innovation. Given the perceived importance of such innovation and technology-driven change in economies today, STEM has become a recurring topic among policymakers worldwide (Croak, 2018).

1.2 Rationale of the Study

The fields of science, technology, engineering, and medicine play a pivotal role in shaping the trajectory of national development and global competitiveness in the

technologically driven economy (Goan & Cunningham, 2006). The disciplines closely related to STEM programs have been integrated into the curriculum and are practiced at the school level in Nepal (CDC, 2007). In response to the need for a qualified workforce, the GoN has invested in STEM education through the University Grants Commission, which supports extensive research and teaching in STEM programs.

Researchers have suggested that success in science and mathematics-related subjects are deeply rooted in interrelated variables like student's ability, attitudes, perceptions, socioeconomic variables, parent and peer influences, school-related variables, and others. Some school-related variables can be named as academic engagement, perceptions, and attitudes (Singh et al., 2022). Since all societies are stratified and the stratification influences educational inequalities, understanding the relationship between the main dimensions of stratification in the larger society and educational phenomena is the key to understanding educational phenomena (Sianou, Kyrgiou, 2010). In Nepali society, caste, gender, and economic status are the three central dimensions of social stratification (Bista, 1991).

A record number of students have been leaving Nepal to pursue education abroad and for better employment opportunities. According to the Ministry of Education, over 110,000 NOCs were issued in the fiscal year 2022/2023. The total education expense for foreign education amounted to NRs. 100.42 billion (NRB, 2023). The education sector of Nepal observed a meagre 4.07 percent growth in the year 2022 (NSO, 2023). In fact, the education exodus has grown so much that most of the prominent colleges in Nepal have started observing unoccupied seats in their most sought-after degrees (Ghimire, 2023).

The number of students studying science, technology, engineering, and medicine has increased significantly in recent years. Although this rise bodes well for the nation's technological progress, a crucial question remains: What influences students to choose STEM? The available data indicates that enrollments are trending positively; however, the questions regarding why it is positive and how we can maintain this interest in STEM still need to be answered. A major research gap exists since there is a dearth of research on the sociocultural, educational, and economic aspects that influence Nepalese students' choices to major in STEM fields. In order to close this gap,

this study looks at the interactions between several factors that either support or undermine students' decision to major in STEM in Nepal.

The results of this study will help us understand what influences the choice of STEM education in Nepal and what interventions can be introduced to promote it. Determining the elements influencing students' decisions will be helpful in formulating policies that promote fairness, inclusivity, and sustained interest in STEM subjects, in addition to addressing the present enrolment trend. The effectiveness of educational efforts and interventions is limited by our need to understand how economic conditions, education experiences, and socioeconomic norms influence students' decisions. As a result, this study will shed light on how socioeconomic status and individual experiences influence STEM major choice.

1.3 Objectives of the Study

The primary objective of this project work is to systematically examine and identify the key factors influencing student's choices of Science, Technology, Engineering, and Medicine (STEM) majors in Nepal. Through a comprehensive analysis of socio-cultural, economic and personal dimensions, the study aims to:

- Examine the role of socio-economic factors like gender, ethnicity and income in selection of a STEM degree.
- Assess the impact of educational experiences, academic scores and family background in students' decision regarding their STEM major.

CHAPTER II

LITERATURE REVIEW

While there is a large and substantial body of research on the factors influencing students' choice of STEM disciplines worldwide, there is a noticeable void in the literature on the subject in Nepal. Research on worldwide trends and Western education systems has dominated existing publications, leaving a large research gap in our knowledge of the particular socio-cultural, educational, and economic aspects influencing STEM major choices in Nepal.

2.1 Theoretical Framework

Social Cognitive Career Theory (SCCT) is derived from Albert Badura's Self-Efficacy Theory and General Social Cognitive Theory (Lent, 2013). SCCT argues that an individual's career path is rooted in the interaction between multiple career elements (Lent et al., 1994). The general social cognitive theory hypothesizes that people are the product of a dynamic interaction between external environmental factors, internal subjectivity factors, and past and present behavior (Bandura, 1986). Factors like personal performance accomplishments, vicarious learning, social persuasion, and physiological states are major factors that affect self-efficacy (Bandura, 1997). Contextual variables influence individuals' career interests and choices by shaping learning experiences in SCCT. The contextual variables often include background contextual affordance, which helps people develop interests, and contextual influences proximal to choose behavior that affects career choice behavior (Lent et al., 1994).

Holland's Theory of Career Choice postulates that personalities and occupational environments can be classified into six different categories, including Realistic, Investigative, Artistic, Social, Enterprising, and Conventional. As such, individuals often search for an environment to express their interests, abilities, and values (Holland, 1985). People prefer jobs where they can be around people who share similar traits. They search for environments that will let them use their skills and abilities and express their attitudes and values, which helps them boost their inherent happiness and fulfilment. Based on the aforementioned six traits, realistic and investigative features are the most relevant to STEM careers (Chen & Simpson, 2015).

Super's theory explains that a career choice is an aggregate of several stages in an individual's lifespan. Namely, the several stages of career development are growth, exploration, establishment, maintenance, and decline. Based on this theory, the early years of life are associated with growth, where individuals will highlight their interests and abilities by engaging in different social activities. The second stage of exploration is associated with the mid-teens to early twenties. Super's theory argues that students can be encouraged to pursue a STEM career with positive encouragement and a supportive environment by nurturing their interests and abilities in this explorative stage (Halim et al., 2018).

This study has adopted the Social Cognitive Career Theory as it appears in the aforementioned literature; environmental factors, STEM self-efficacy, perception of STEM careers, and interest in STEM are four crucial factors affecting the choice of a STEM major.

2.2 Empirical Framework

According to Vu et al. (2019), students were likelier to choose STEM majors if they performed well academically in STEM-related disciplines. The study also discovered that factors influencing high school student's decision to major in STEM in college included gender in addition to race, and/or ethnicity. Additionally, they maintain that choosing a major based on intrinsic goals might help students avoid changing their major and result in greater success, happiness, and contentment. Instructors, friends, and family must recognize the impact role models have on students' thoughts and preconceptions about particular disciplines and how they affect their choice of STEM major.

Halim et al. (2018) found that environmental sub-factors like classroom activities, media influences, and non-classroom activities influence an individual's self-efficacy and perception of STEM careers. The interaction between these variables influences their interest in STEM disciplines.

As per Niu's (2017) research, minorities were more likely than the majority to select STEM as their undergraduate major if other criteria were equal, even if they were underrepresented in STEM disciplines. Enrolment in STEM programs has been significantly predicted by pre-college mathematical performance. However, this was not in line with the minorities. Students' mathematics performance became stronger

with an increase in the family's socioeconomic status. Students with higher mathematics scores have a higher chance of succeeding in STEM majors, while those with poor mathematics preparation are likelier to leave STEM fields. The socioeconomic status of the student's household was found to be positively connected with their level of mathematics preparation. The socioeconomic level of a student's household improves with increased mathematics proficiency.

Wang (2013) mentions that despite much prior literature focused on mathematics achievement when examining the influence of high school experience on student interest and entrance into STEM fields, however, the effect of students' exposure to mathematics and science courses is even stronger than that of mathematics achievement. An earlier introduction and exposure to math and science-related courses could be an effective method to boost students' interest in pursuing STEM fields of study. This means students' interest in pursuing STEM could be evoked in response to direct exposure to these courses. They also pointed out that racial backgrounds still largely impact the way in which high school math and science learning is linked to STEM aspirations. The effect of high school exposure to math and science courses on STEM intent is heterogeneous, accruing least to underrepresented minorities.

Hutchinson-Anderson et al. (2015) explores the role of middle school and high school teachers in students' participation in STEM fields of study. The study found that the students who felt prepared theoretically, in mathematics and other science subjects, and in the usage of practical lab instruments had a teacher who was thoroughly interested in the subject matter. Students also felt more prepared in STEM majors when their teachers engaged the students and helped them to see connections outside of the classroom.

Tyler-Wood et al. (2018) investigated the gender-wise difference in the attitude of middle school students towards STEM-related subjects in the US. The research suggested that science, engineering, and mathematics interests were statistically significant variables in explaining the attitude toward STEM for both boys and girls. However, understanding a solenoid with the aforementioned variables explained 47% variance in boys pursuing a STEM career. Similarly, for girls, the addition of technological understanding explained 36% of the variance in pursuing a STEM career.

Goel et al. (2018) found that medicine majors from high-income and middle-income countries were motivated to pursue the degree due to their interest in science, flexible work hours, and independence. Societal factors like prestige, job security, and financial security were found to be statistically significant variables in choosing medicine as their major in higher and middle-income countries. Additionally, humanitarian factors like serving the poor and underprivileged also explain the variation in interest in medical studies. This aligns with Maslow's hierarchy of needs, a theory of motivation.

Bashir et al. (2023) found that in low-income countries like Sudan, self-interest was the most common factor influencing the choice of a medical major. Getting a high score in high school also motivated students to pursue medicine as their career. However, parental and peer pressure were observed in students pursuing medicine majors in low-income countries, contrary to those in high or middle-income countries. Furthermore, the societal prestige that comes along with medical degrees also plays a part in students choosing medicine as their major in low-income countries.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Materials

3.1.1 Survey Instrument

The survey instrument utilized in this study was a structured questionnaire administered through Google Forms. The questionnaire was designed to gather insights into students' perceptions, attitudes, and motivations towards STEM education in Nepal. It comprised of a series of questions addressing various aspects such as reasons for choosing a specific STEM degree, socio-economic background, intrinsic motivation and prior educational experiences. The design of the questionnaire was informed by existing literature on STEM education and tailored to the Nepalese context to ensure relevance and comprehensiveness.

3.1.2 Sample Size

The survey collected data on 55 individuals. These 55 observations serve as prior distribution to model a posterior distribution for variables of interest to generate data points given the parameters and distribution. Using Markov Chain Monte Carlo (MCMC) computational technique, we generated exactly 10000 observations, with each chain generating 2000 observations.

For the MCMC diagnostic, we performed the Gelman-Rubin convergence diagnostic, where the potential scale reduction factors (PSRF) for all variables were close to 1, indicating no convergence issues were detected. We also conducted the Heidelberger-Welch Diagnostic Test, and all generated values passed the stationarity test and halfwidth mean test, indicating stable chains and accurate mean estimate. Trace and autocorrelation plots were generated for all generated values, and the plot showed well-mixed chains. The autocorrelation plots indicated low autocorrelation, supporting the convergence diagnostics.

3.1.3 Statistical Analysis Software

Data analysis was conducted using the R programming language, leveraging the Rstan package for Bayesian statistical modelling. Rstan provided a comprehensive suite of tools for fitting complex statistical models to survey data, allowing for a robust

inference and interpretation of results. The use of Rstan facilitated the implementation of sophisticated Bayesian method, Markov Chain Monte Carlo (MCMC), tailored to the specific research questions and objectives of the study.

Similarly, the nnet library was also utilized in order to run a multinomial logistic regression. The library provided much needed tools for fitting the data into a regression model and also with the change of base of comparison for the multinomial regression to understand different levels of significance of variables when different level is held as base level.

3.2 Methods

3.2.1 Study Design

This research adopts a mixed-methods approach to explore the multifaceted factors influencing students' choice of STEM major in Nepal. The study begins with a qualitative phase, employing a cross-sectional survey design to collect data from a sample of undergraduate students across different educational institutions offering STEM majors in Nepal. The survey instrument, a questionnaire, was developed using Google Forms and aimed to capture the demographic information about the representative sample to contextualize the findings and explain the narrative stemming from the sample data.

Following the qualitative phase, the study incorporates a quantitative element to delve deeper into students' decision-making processes. These qualitative variables are necessary to model our statistical analysis to understand how these socioeconomic and academic variables affect the students' decision-making process regarding their choice of STEM major.

By integrating both quantitative and qualitative methods, the mixed-methods approach enables a comprehensive and integrated understanding of the composite relationship between the variables impacting students' decision regarding their major. The use of qualitative and quantitative variables enhances the validity and reliability of the findings, allowing for a comprehensive analysis of the factors driving student engagement with STEM education and informing targeted interventions to promote greater participation in STEM fields in Nepal.

3.2.2 Data Collection

Data for this study were collected through a structured questionnaire administered to undergraduate students in Nepal. The questionnaire was designed to capture a range of variables related to students' perceptions, attitudes, and motivations toward STEM education, as well as demographic information. The questionnaire was divided into sections: the first contained questions regarding general demographic information, and the second dealt with students' intrinsic motivation for their significant—similarly, the third section dealt with financial matters like tuition, scholarships, additional costs, etc. The last section asked about their past schooling, secondary science, and math scores, and whether or not their high school offered STEM-related extracurriculars.

The questionnaire was carefully designed to ensure clarity, comprehensiveness, and relevance to the Nepalese context. Before data collection, a pilot study assessed the questionnaire's readability, comprehensibility, and appropriateness for the target population. Based on feedback from the pilot survey, minor revisions were made to improve the clarity and structure of the questionnaire.

The questionnaire was then distributed to undergraduate students in Nepal through different STEM departments in different colleges and educational institutions. A stratified random sampling approach was employed to ensure representation from different STEM majors across various institutions.

After collecting the survey responses from undergraduate students in Nepal, Markov Chain Monte Carlo (MCMC) methods were employed to generate additional samples from the observed data. By leveraging MCMC, additional samples were generated to augment the original dataset, enabling a more robust analysis of the relationships between predictor variables and the choice of STEM majors. This approach helped to address potential limitations associated with finite sample sizes and provided insights into the variability of model estimates under different sampling scenarios.

The predicted data was generated using the No-U-Turn Sampler (NUTS), an extension to Hamiltonian Monte Carlo (HMC). NUTS uses a recursive algorithm to build a set of likely candidate points that spans a wide swath of the target distribution,

spotting automatically when it starts to double back and retrace its steps (Gelman & Hoffman, 2014).

The variables of interest for this research were a combination of discrete and continuous variables. The continuous variables were rather naturally occurring ones; therefore, we used normal distribution to model them.

$$f(x;\ \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Similarly, we used the Dirichlet distribution to model the discrete variables of interest. The Dirichlet distribution in a multivariate generalization of beta distribution. In Bayesian analyses, the Dirichlet distribution is often used as a prior distribution of the parameters of the multinomial distribution (International Encyclopaedia of Education, 2023). Here, x_i is the probability associated with a category with sample size of α_i, $i = 1, \dots, n$.

$$p(x_i|\alpha_i) = \frac{(\alpha_i - 1)!}{(\alpha_1 - 1)! \times \dots \times (\alpha_n - 1)!} \prod_{i=1}^{n} x_i^{\alpha_i - 1}$$

3.2.3 Data Analysis

In this study, a comprehensive data analysis approach was employed to investigate the factors influencing students' choice of STEM majors in Nepal. The analysis began with a descriptive examination of the survey responses, providing insights into the demographic characteristics of the study population, as well as key trends and patterns observed in students' perceptions and motivations towards STEM education. Summary statistics was utilized to characterize the dataset and identify potential relationships between variables of interest.

In our data analysis, income, educational expenses, scholarship amounts are measured in Nepalese Rupees (NRs). These financial variables provide insights into the economic background against which educational choices are made. Additionally, we assessed academic performance through variables such as GPA, mathematics grade, and science grade, each ranging from 1.6 to 4, representing students' proficiency in these subjects. Our analysis incorporates discrete variables to capture categorical attributes such as ethnicity, college type, university, permanent address, province, degree, parental expectations, peer influence, intrinsic motivation, social reputation,

and house type. Each discrete variable assumes a categorical value that reflects different demographic, socio-economic, and personal characteristics, facilitating a sound understanding of the diverse influences on educational decision-making. The dummy variables are used to capture categorical attributes such as gender, first-generation college status, relocation status, government funding, prior stem member status and STEM activity involvement. These dummy variables allow us to analyse the said effect of variables on students' academic endeavours by acting as binary indicators that reflect particular categories within each attribute.

Chi-square test was performed to analyse whether gender played a role in choice of major in the STEM field. The data consisted of number of male and female students in each field, Science, Technology, Engineering and Mathematics. The chi-square test statistic (χ^2) was calculated as:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where,

O_i= Observed frequency of each category.

E_i= Expected frequency of each category.

The degrees of freedom (df) for chi-square test was calculated as:

$$df = (r - 1) \times (c - 1)$$

where,

r = number of rows

c = number of columns

A one-way Analysis of Variance (one-way ANOVA) was conducted to check whether income status had any impact on the choice of major among STEM degrees. The data consisted of the income level of students' families in each field.

Multinomial logistic regression was subsequently employed to explore the associations between predictor variables and the outcome variable of STEM major choice. The regression model was constructed to accommodate the categorical nature of the outcome variable, which encompassed multiple levels representing different

STEM majors available to students. There are four different categories in STEM: one represents science, two represent technology, three represent Engineering, and four represent medicine. Predictor variables included demographic factors, motivational factors, and prior educational experiences. The model specification was guided by relevant literature and theoretical frameworks, carefully considering variable selection and operational definitions to ensure model parsimony and interpretability.

A multinomial logistic regression model is a logistic model generalized to categorical variables Y with more than two possible levels or outcomes, namely $\{1, \dots, J\}$. Given the predictors, $X_1, \dots, X_p$, multinomial logistic regression models the probability of each level j of Y by,

$$p_j(x) = \mathbb{P}[Y = j | X_1 = x_1, \dots, X_p = x_p]$$

$$= \frac{e^{\beta_{0j} + \beta_{1j} X_1 + \cdots + \beta_{pj} X_p}}{1 + \sum_{l=1}^{J-1} e^{\beta_{0l} + \beta_{1l} X_1 + \cdots + \beta_{pl} X_p}}$$

For $j = 1, \dots, J - 1$ and (for the last level of J),

$$p_j(x) = \mathbb{P}[Y = j | X_1 = x_1, \dots, X_p = x_p]$$

$$= \frac{1}{1 + \sum_{l=1}^{J-1} e^{\beta_{0l} + \beta_{1l} X_1 + \cdots + \beta_{pl} X_p}}$$

The above equations imply that $\sum_{j=1}^{J} p_j(x) = 1$ and that there are $(J - 1) \times (p + 1)$ coefficients. Taking the quotient between the above equations and applying logarithm, we have,

$$\log\left(\frac{p_j(x)}{p_J(x)}\right) = \beta_{0j} + \beta_{1j} X_1 + \cdots + \beta_{pj} X_p$$

Here, if $\beta_{0j} > 0$, then $Y = j$ becomes more likely than $Y = J$ when $X_1 = \cdots = X_p = 0$. Equivalently, if $\beta_{0j} < 0$, then $Y = j$ becomes less likely than $Y = J$ when $X_1 = \cdots = X_p = 0$.

Similarly, if $\beta_{lj} > 0$, then $Y = j$ becomes more likely than $Y = J$ for unit increase in X_j. Likewise, if $\beta_{lj} < 0$, then $Y = j$ becomes less likely than $Y = J$ for unit increase in X_j. This holds if and only if the remaining variables $X_1, \dots, X_{l-1}, X_{l+1}, \dots, X_p$ do not change.

The generated data from MCMC were incorporated into the multinomial logistic regression analysis, alongside the observed survey responses, to refine the estimation of model parameters and enhance the reliability of the results. Convergence diagnostics and sensitivity analyses were performed to assess the stability of the MCMC chains and ensure the validity of the generated samples.

The interpretation of results from the multinomial logistic regression analysis focused on identifying significant predictors for STEM major choice and understanding their implications for informing educational policies and interventions in Nepal. Model diagnostics and sensitivity analyses were used to evaluate the validity and reliability of the regression results, while acknowledging potential limitations and assumptions inherent in the analysis.

CHAPTER IV

RESULTS AND DISCUSSION

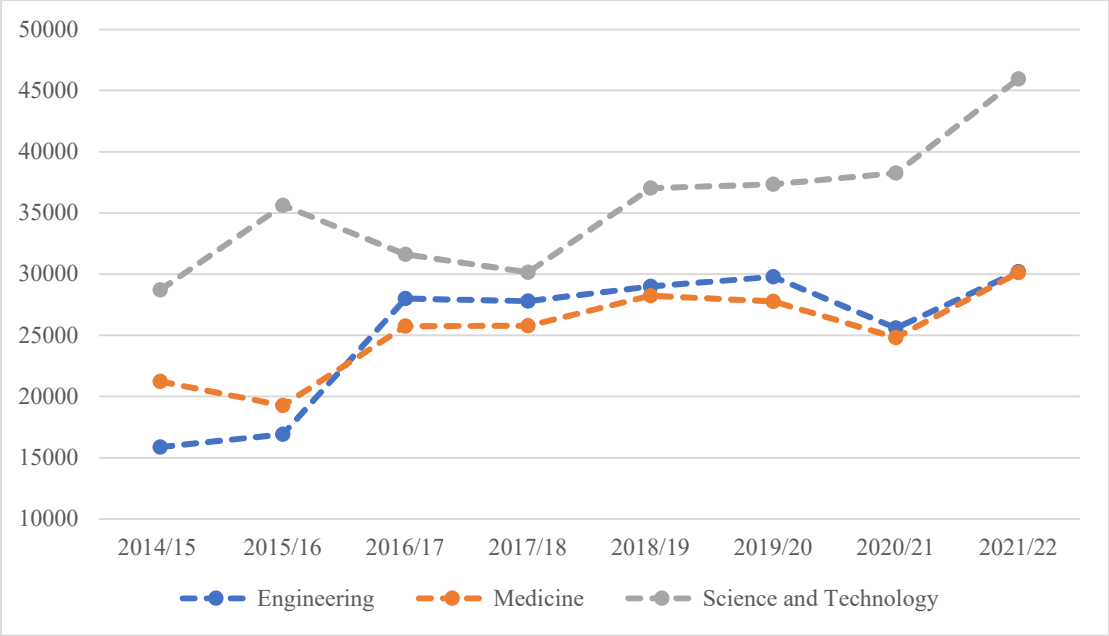
In this chapter, we present description regarding the data and the findings of the Chi-Square test, One way Analysis of Variance (ANOVA) and multinomial logistic regression analysis, exploring the factors influencing students' choice of STEM major in Nepal, with Science as the reference category, due to it having the smallest number of observations.

4.1 Descriptive Statistics

STEM has always been a coveted education in Nepal, deemed prestigious by the society due to its better career opportunities. Around 44616 students enrolled in STEM majors in year 2015, and now that number has increased to 76171 in 2022 (University Grants Commission, 2015; 2022). The average cost of higher education in public institutions per student in 2021 was NPR 64,170 compared to NPR 135,000 in private institution (Ministry of Education, Science and Technology, 2021). Therefore, given the GDP per capita of Nepal is USD 1336, STEM studies are quite the luxury in Nepal (World Bank, 2022).

Figure 1

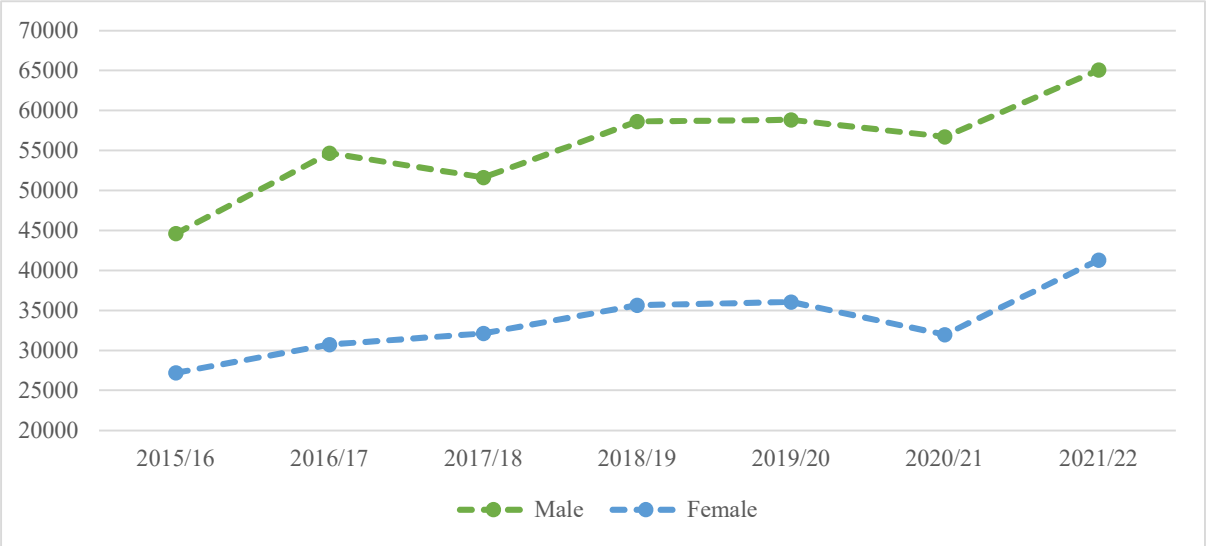
Number of Student Enrollments in STEM majors in Nepal



While the number of students pursuing a STEM degree in Nepal has seen an overall positive trend, the engineering, science and technology fraction is dominated by male. In the academic year 2021/22, 79% of students enrolled in Engineering, 62% in Science and Technology were male. With 60% of enrolled students being female, medicine continues to be the only field where females outnumber males. In Nepal, there are 143 institutions offering courses connected to science and technology, 85 and 54 campuses offering medicine and engineering related courses respectively (UGC, 2022).

Figure 2

Number of male and female enrolled in STEM major in Nepal

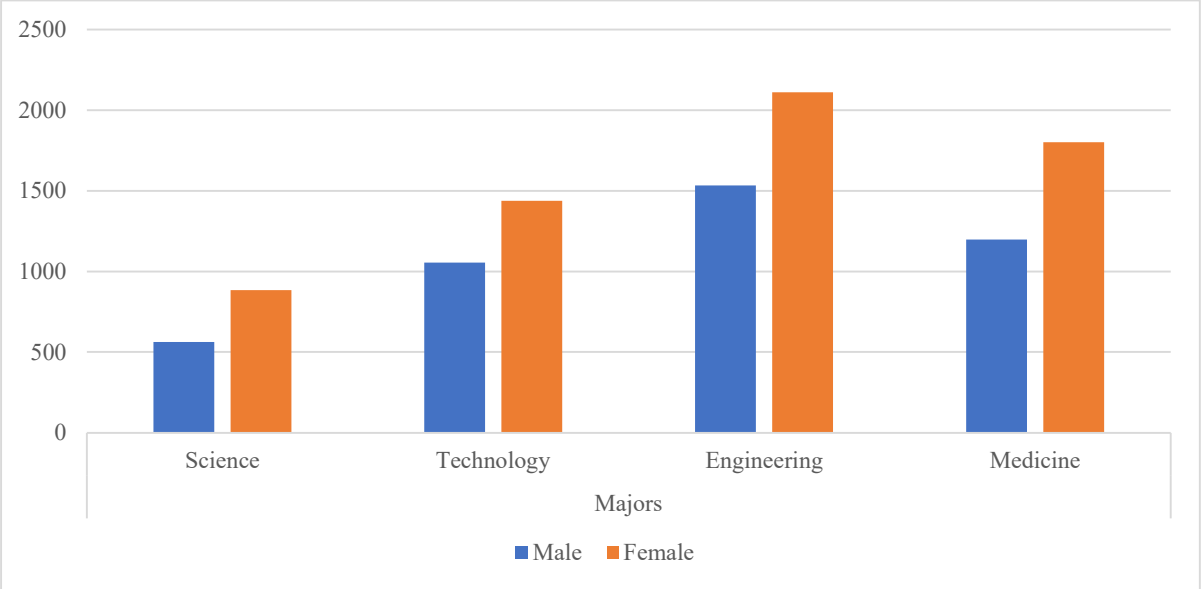


From the 55 observations collected through a survey, we built a MCMC model and generated 10000 observations with one chain generating 2000 observations each. The bar graph below shows the number of male and female in respective STEM field from the generated observations.

There was total 4063 male students and 5947 female students in the generated posterior distribution sample. As we can see, there are more female students than male students as per the observations generated by the MCMC model, that is because the prior distribution had more female students than male students and females are more likely to respond to online surveys than males (Smith, 2008).

Figure 3

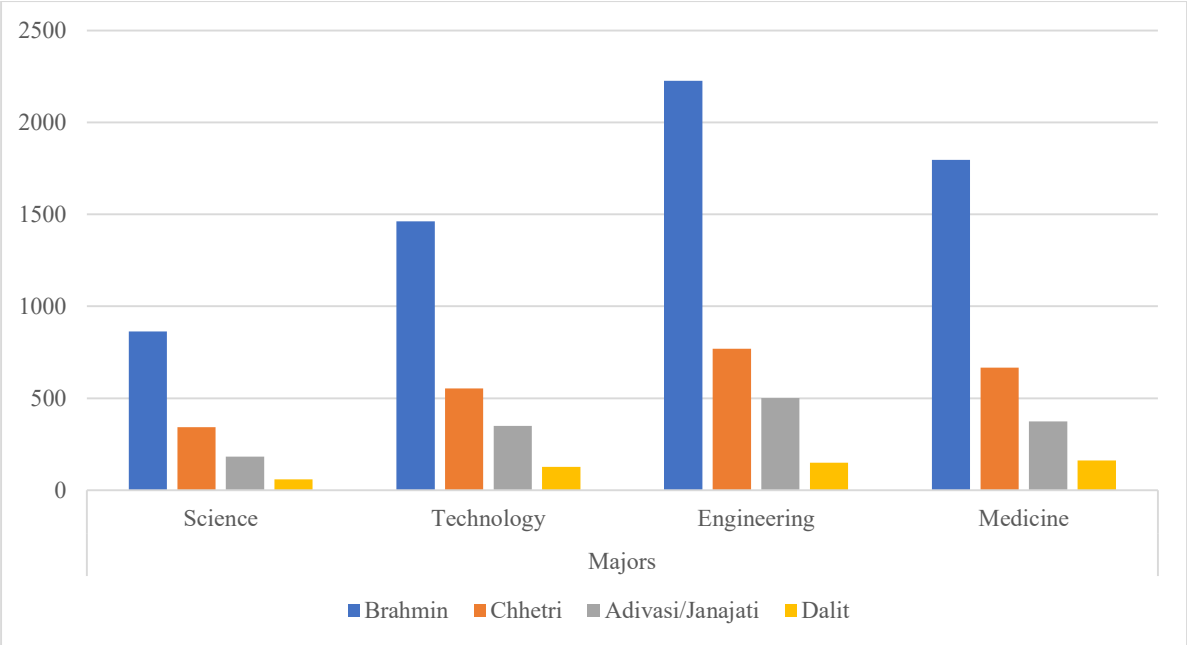
Gender-wise number of students in different majors in the sample



For ethnicity, there were only four reported ethnicities in the prior distribution, Brahmin, Chhetri, Adivasi/Janajati and Dalit. The students belonging to Brahmin and Chhetri groups had the highest representation, while the other two groups had minimum representation.

Figure 4

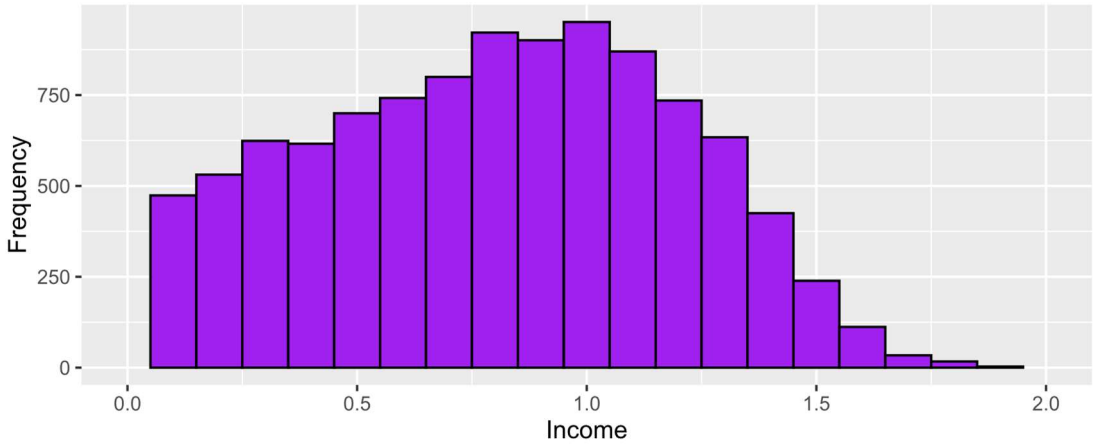
Ethnic Composition of STEM Majors in sample



As suggested by the histogram below, we can say that the income variable approximately follows normal distribution with skewness to the right. The majority of the income values are distributed towards the mean with few sample values in the higher income range.

Figure 5

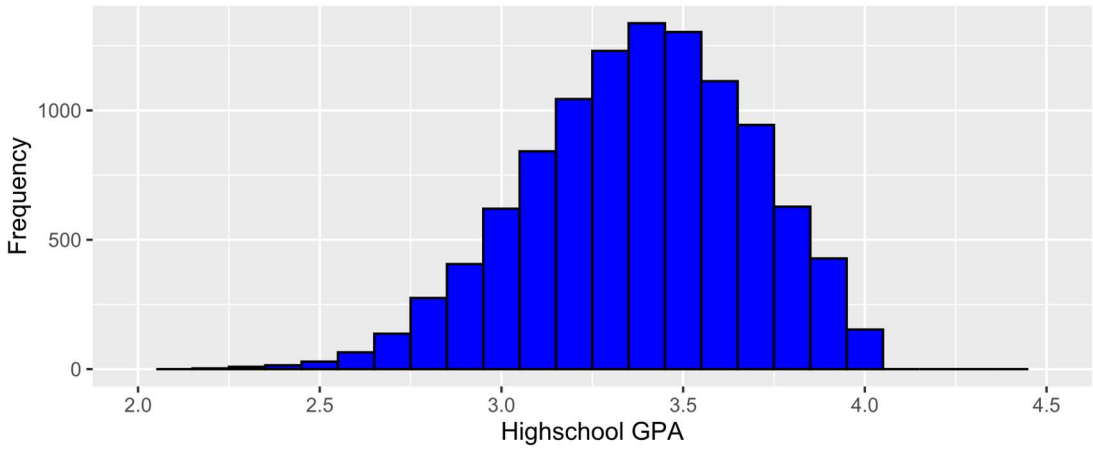
Histogram of Income Variable



The histogram of high school GPA suggests that the variable is normally distributed with most values centred around the mean and less observations in the lower GPA range.

Figure 6

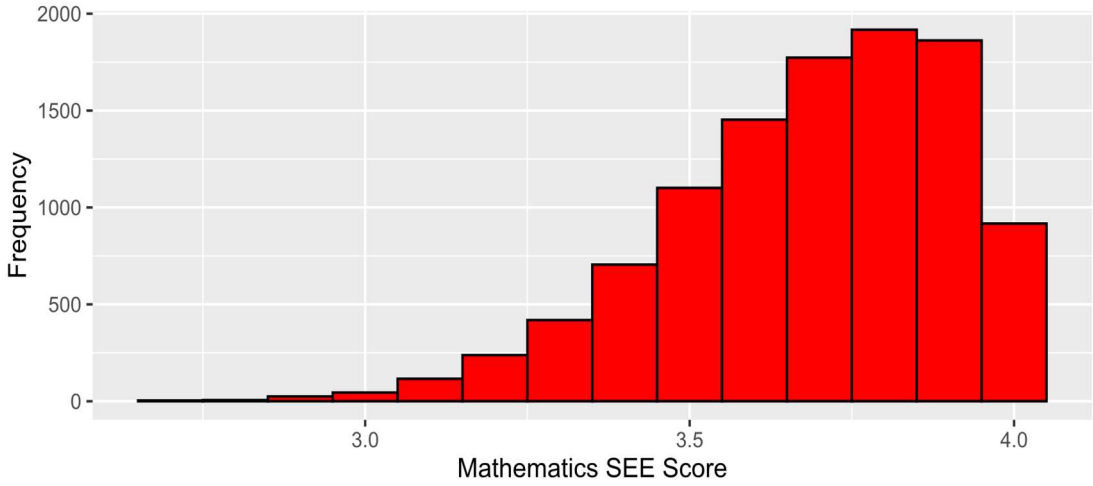
Histogram of Highschool GPA



The histogram of mathematics score shows that the variable approximately follows normal distribution and is skewed to the left. The distribution shows that majority of the observations are distributed around the mean value.

Figure 7

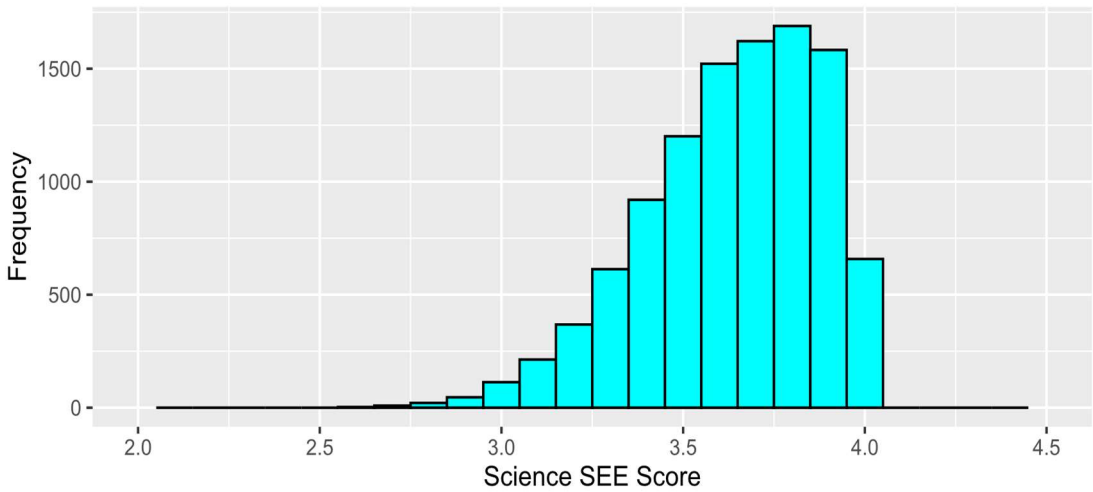
Histogram of Mathematics Score in SEE



The histogram below shows the distribution of Science SEE score of the respondents in the sample. The variable approximately follows normal distribution with skewness to the left. The majority of the observations are distributed around the mean with very few in the lower GPA side.

Figure 8

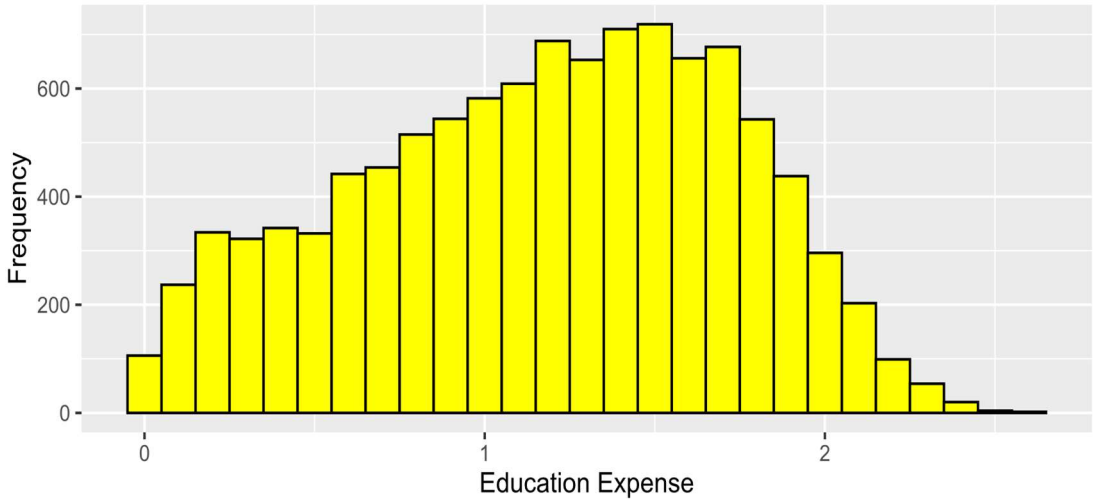
Histogram of Science Score in SEE



The histogram below shows the distribution of education expense of the sample observations. The shape of histogram is a perfect bell shape suggesting a normal distribution with most values centred about the mean.

Figure 9

Histogram of Education Expense



The histogram of scholarship suggests that there is a skewed distribution with majority of the students not receiving any kind of scholarship, only a select few are awarded substantial financial aid.

Figure 10

Histogram of Scholarship

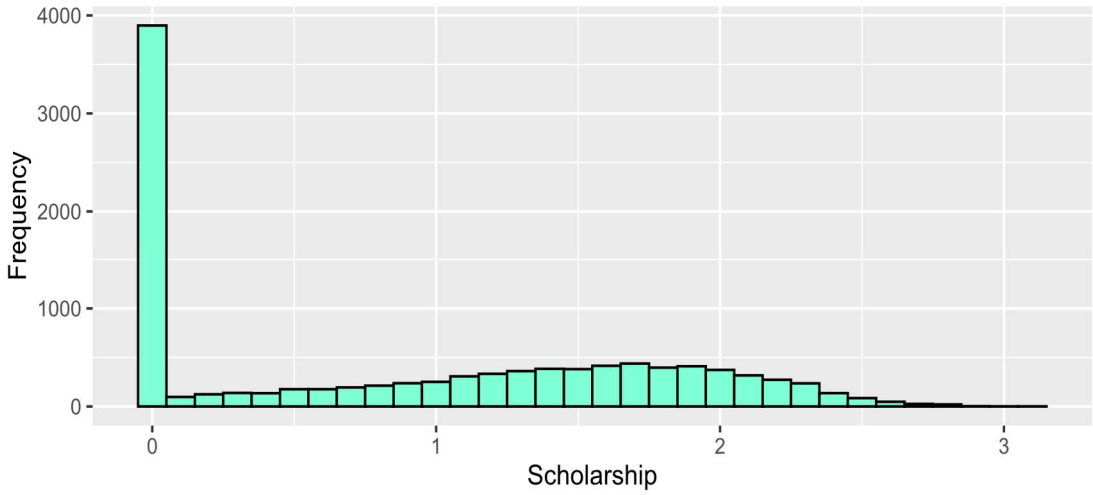


Table 1

Descriptive Statistics of the Discrete Variables

Variable	Category	Count	Percentage
Gender	Male	4063	40.63%
	Female	5947	59.47%
Degree	Science	1367	13.67%
	Technology	2354	23.54%
	Engineering	3444	34.44%
	Medicine	2835	28.35%
Ethnicity	Brahmin	5997	59.97%
	Chhetri	2203	22.03%
	Adivasi/Janajati	1331	13.31%
	Dalit	469	4.69%
First Generation College Student	Yes	4497	44.97%
	No	5503	55.03%
Prior STEM Member	Yes	4211	42.11%
	No	5789	57.89%
Government Funding	Yes	2021	20.21%
	No	7979	79.79%
STEM Activity in High School	Yes	4840	48.40%
	No	5160	51.60%

4.2 Empirical Results

4.2.1 Analysis of Variance (ANOVA)

In this section, we have conducted a one-way ANOVA to test whether there is difference in means of income belonging to different STEM majors. In other words, we can test whether the difference in income level of families of students in different majors is statistically significant or not. The results showed that there was no significant effect of family income on students' choice of STEM major at 1% level of significance as, $p = 0.25$.

Table 2

ANOVA Results for the Impact of Income on STEM Major Selection

Source of Variation	df	Sum Square	Mean Square	F	p-value
Degree	3	4	1.23	1.37	0.25
Residuals	9996	9532	0.901		
Total	9999	9536			

We also checked whether the GPA of students played a statistically significant role in their choice of STEM major. The results showed that there was significant impact of GPA on students' decision regarding their major at 5% level of significance as $p = 0.03$. Hence, we can conclude that the mean high school GPA of students of different STEM majors is different.

Table 3

ANOVA Results for the Impact of High School GPA on STEM Major Selection

Source of Variation	df	Sum Square	Mean Square	F	p-value
Degree	3	0.8	0.266	2.87	0.0348*
Residuals	9996	980.6	0.092		
Total	9999	981.6			

4.2.2 Chi-Square Test

Chi-Square Test was done to study whether gender played a role in selection of major among STEM majors in Nepal. According to the results, we fail to reject the null hypothesis at 5% level of significance as, $p = 0.058$. Given the rejection margin is so slim, we can deduce that there is a borderline association between gender and selection of STEM major.

Similarly, to understand whether ethnicity plays a role in selection of STEM major, we again conducted a Chi-Square Test. The results show that there is marginal association between ethnicity and decision regarding STEM major in Nepal, $p = 0.059$.

4.2.3 Multinomial Logistic Regression

The multinomial logistic regression was run, with degree as the dependent variable, on the posterior distribution generated using the MCMC method. Variables like Gender, First Generation, Stem Activity, Government Funding, Relocation, and Prior STEM Member are modelled as dummy variables. Income, Education Expenses, and Scholarship are all measured in Nepalese Rupees (in lakhs) and are continuous variables along with GPA, Maths Score, and Science Score, which account for high school GPA and SEE grades for mathematics and science. The remaining variables, like Parental Expectations, Intrinsic Motivation, and others, are discrete variables.

The regression table presented below gives us the probability percentage of selecting the specified major over science, for unit increase in said variable. The science major was chosen as the reference level because it had the lowest number of respondents. The standard error refers to the standard deviation of the estimate from the regression line. The intercept represents the log odds of choosing the given major over science when all the dependent variables are zero. The positive coefficients represent an increase in log odds, while the negative coefficients represent a decrease in log odds of choosing the specified major over science.

For the validity of the model, we produced the correlation matrix to check whether any of the variables are highly correlated with each other or not. We observed no variables had a correlation of more than 0.2. The histogram of deviance residuals showed that they are normally distributed. We also conducted a overdispersion test, the resulted value was 0.725 suggesting that variability in less than expected indicating the data fits the data without overdispersion concerns.

The intercept for all three majors, Technology, Engineering, and Medicine, indicate that on baseline a student is more likely to choose any of these three majors over science. The income variable indicates that the probability of choosing engineering and medicine over science increases with income, however, decreases for technology. In other words, in higher-income families, there is less likelihood of students choosing technology over science. This can be attributed to the fact that high-income families often encourage their children to pursue stable and traditional careers (Martjin et al., 2020). Education expenditure is not statistically significant in impacting students' decisions regarding their career paths because the perceived education cost doesn't

always directly influence students' choice of education field. This is because most families don't consider the immediate cost but the long-term returns on investment, and fields like technology, engineering, and medicine have always been high-earning fields (Baum et al., 2016).

Being a first-generation college student has higher probability of choosing science over other majors because a person who is a first-generation college student tends to choose those that are less costly and are perceived as stable careers (Pascarella et al., 2004). Being a female increases the probability of choosing technology, engineering, or medicine over science. This is due to the fact that there are more female respondents to this survey and given that posterior distribution was modelled using that data; however, in medicine, the probability is a very marginally higher that 50% which can be attributed to the fact that medicine has a pretty even distribution of students (gender-wise). Receiving government funding increases the probability of pursuing technology, engineering, or medicine over science. However, the magnitude of probability is smaller for technology compared to the other two. This is because engineering and medicine are highly coveted degrees that are very costly to pursue; therefore, getting any kind of government funding, like scholarships or financial aid, can help students pursue costly career paths (Castleman et al., 2016).

Increase in parental expectations, peer influence, and intrinsic motivation all increase the probability of choosing a particular major over science. Higher parental expectations can make students choose degrees that are more lucrative and are perceived as prestigious (Fan et al., 2010). Intrinsic motivation can drive students to engage in activities that give them inherent satisfaction; therefore, students who enjoy innovation and problem-solving will choose either technology or engineering as their major. Those who are interested in medicine and human anatomy will be choosing medicine as their major (Ryan & Deci, 2000). With higher level of education for parents, the children are more likely to choose engineering or medicine. This is due to high income families generally tend to have parents who have a strong education background propelling students to choose degrees which lead to a stable career path.

Table 3

Multinomial Logistic Regression Coefficient Table

	Technology			Engineering			Medicine		
	Probability (%)	Std Error		Probability (%)	Std Error		Probability (%)	Std Error	
Intercept	92.38	0.00	***	88.47	0.00	***	88.56	0.00	***
Income	48.47	0.00	***	59.20	0.00	***	58.51	0.00	***
Education Expense	50.00	0.00		50.00	0.00		50.00	0.00	
Scholarship	50.12	0.00	***	49.77	0.00	***	49.81	0.00	***
GPA	45.59	0.00	***	43.51	0.00	***	43.02	0.00	***
Maths Score	45.97	0.00	***	45.92	0.00	***	48.70	0.00	***
Science Score	43.35	0.00	***	51.41	0.00	***.	48.30	0.00	***
Gender	53.56	0.00	***	53.34	0.00	***	51.25	0.00	***
First Generation	49.23	0.00	***	46.90	0.00	***	47.66	0.00	***
Parental Expectations	50.33	0.00	***	49.80	0.00	***	50.20	0.00	***
Peer Influence	50.28	0.00	***	51.04	0.00	***	50.25	0.00	***
Intrinsic Motivation	50.59	0.00	***	50.51	0.00	***	50.06	0.00	***
Social Reputation	49.95	0.00	***	49.12	0.00	***	49.41	0.00	***
Prior STEM Member	48.77	0.00	***	49.05	0.00	***	49.98	0.00	***
Relocation	49.96	0.00	***	49.19	0.00	***	47.68	0.00	***
Government Funding	51.27	0.00	***	52.72	0.00	***	55.07	0.00	***
Stem Activity	49.54	0.00	***	49.72	0.00	***	49.59	0.00	***
Father Education	50.70	0.00	***	50.97	0.00	***	51.26	0.00	***
Mother Education	49.21	0.00	***	50.05	0.00	***	50.05	0.00	***

$McFadden's\ R^2 = 0.029$

Significance codes: ‘***’ 0 ‘**’ 0.01 ‘*’ 0.05 ‘,’ 0.1 ‘ ’ 1

Academic factors like high school GPA, mathematics score and science score, all have probabilities less than 50%, implying the higher these academic metrics the students are more likely to choose science over other majors. This can be attributed to students who score better in mathematics and science are more interested in core science and mathematics and have an inclination towards research-oriented careers (Maltese et al. 2011).

Having someone in your family who has already pursued a STEM major can also help students shape their decision regarding their major. In this case, there is low probability of choosing technology or engineering as your major over science. However, this has very minimal to no impact on students' choice of pursuing medicine over science. Given the students have someone in their family who has pursued technology or engineering, the students can better understand the rigors and potential job market issues of pursuing Technology or Engineering fields, which can drive them to choose science programs over them (NSF, 2020). However, for medicine, family exposure to STEM may increase their familiarity with technical fields. However, medicine generally requires personal motivation and experiences beyond family influence (AAMC, 2020).

CHAPTER V

CONCLUSION AND RECOMMENDATION

5.1 Conclusions

In conclusion, our study provides valuable insights into the different dynamics that play a role in students' decisions regarding their STEM major choice in Nepal. Through a comprehensive multinomial logistic regression model, we have identified a range of different factors that significantly impact students' decision for their career path in Science, Technology, Engineering and Medicine (STEM).

Our findings highlight the significance of various socio-economic, academic, and personal factors in shaping students' choice of STEM major in Nepal. Factors such as first-generation college student status, parental expectations, peer influence, GPA, and government funding appear as key factors influencing students' decision regarding their STEM major in Nepal.

In addition, our research sheds light on different level of impacts of various variables across different degree choices, providing necessary insights that shape students' decision regarding their education and career path. While financial aspects don't play a huge role in decision of students' regarding their degree, factors such as gender, ethnicity, GPA, and government funding have a pivotal role in shaping their decision regarding their selection of STEM major.

Overall, our study contributes to a deeper understanding of factors influencing students' selection of STEM major among Nepalese students, offering insights for policymakers, educators and stakeholders seeking to promote equitable access to STEM education which is closely linked with economic growth of a country. By addressing different needs and bring interventions plans and policies, the government can foster an environment of more inclusive and diverse STEM workforce in Nepal.

5.2 Recommendations

Based on our findings, here are some of the recommendations proposed to enhance STEM education and promote more informed decision-making among Nepalese students,

1. Introduction of career counselling services within high schools and universities, which can help students explore their interest and career aspirations and provide guidance about different STEM career paths at the same time (Whiston and Quinby, 2009).
2. Organizing fellowship programs and interaction programs with the industry which can offer students a valuable insight into the industry they are studying to be a part of. The students can have someone they look up to as an inspiration in their field which can give a direction to their passion (Crisp and Cruz, 2009).
3. Recognizing the impact of socio-economics factors on education opportunities and introduction of policies which can bring equitable access to education. Scholarship programs, aid or grants for students with good academic record but poor financial condition, which can help ease the financial burden (Dynarski and Scott-Clayton, 2013).
4. Integrating STEM-focused curriculum such as project-based learning approaches and interdisciplinary coursework with regular subjects to make education more engaging and relevant for students (Darling-Hammond, 2008).
5. Research is an important part of STEM education; the government should allocate resources for the development of state-of-the-art STEM research facilities like laboratories and fund research projects to promote innovation and improve students' learning experiences (NRC, 2012).
6. Teachers and professors must have interdisciplinary knowledge in various fields and must work with other teachers. They must be provided with professional development opportunities in STEM education which can help them nurture students' critical thinking and problem-solving skills (Desimone, 2009).
7. Revamping the syllabus for STEM education in university level and introduction of subject specific programming in STEM as it has become the backbone of research (Wing, 2006).
8. Collaboration of government with the private sector to increase employment opportunities by creating an entrepreneurial economy which promotes market expansion and helps in increasing employment opportunities and help students direct into STEM fields (Audretsch and Belitski, 2017).

By implementing these recommendations, policymakers and the government can work towards fostering a culture of STEM education in Nepal,

empowering students to pursue their passions and contribute meaningfully to advancement of STEM fields and country as a whole.

5.3 Limitations of the Study

The project has potential limitations, which are mentioned below in detail.

- This research is done for academic purpose.
- The statistical model employed in the research may have excluded certain associated variables.
- The sampling methods used may have an impact on the findings of the research, given some groups of students are underrepresented or excluded.

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APPENDIX

Questionnaire

Response Number:

Respondent Email:

Date Submitted:

1. Please enter your sex.
    - Male
    - Female
2. Please mention your caste.
    - Brahmin
    - Chhetri
    - Adivasi/Janajati
    - Dalit
    - Others: ……
3. Please indicate what type of college you are currently enrolled in.
    - Public
    - Private
    - Community
4. What university are you currently enrolled in?
    - Tribhuvan University
    - Kathmandu University
    - Purwanchal University
    - Pokhara University
    - Other: …….
5. Which of the following best describes your permanent address?
    - Metropolitan City
    - Sub-Metropolitan City
    - Municipality
    - Rural Municipality
6. Which of the following states are you from?
    - Koshi Pradesh

- Madhesh Pradesh
- Bagmati Pradesh
- Gandaki Pradesh
- Lumbini Pradesh
- Karnali Pradesh
- Sudurpaschim Pradesh

7. Are you a first-generation college student?
    - Yes
    - No
8. Please specify the field within STEM in which you are currently pursuing your degree.
    - Science
    - Technology
    - Engineering
    - Medicine
9. Please select the appropriate degree for each of the following variables that influenced your STEM major decision (1 being the lowest and 5 being the highest).

	1	2	3	4	5
Parental Expec...	○	○	○	○	○
Peer Influence	○	○	○	○	○
Intrinsic Motiv...	○	○	○	○	○
Social Reputati...	○	○	○	○	○

10. Did any family member pursue a STEM major before you?
    - Yes
    - No
11. What is the total average annual income of your household?

……..

12. What is the total amount spent on education-related expenses, encompassing tuition, books, and materials, over the past year?

…….

13. What is the amount of scholarship you've received for your degree?

    …….

14. What type of house does your family own?
    - RC Frame
    - Mud and Stone
    - Bamboo
    - Wooden
    - Others: ……..

15. Did you have to relocate in order to pursue your degree?
    - Yes
    - No

16. What is the distance (in km) between your residence and your educational institution?

    ……..

17. Are you part of any government program funding your study?
    - Yes
    - No

18. Did you have any access to STEM related activities or clubs in high school?
    - Yes
    - No

19. What was your GPA in (+2) high school?

    ……

20. How much did you score in Mathematics in SEE?

    ……

21. How much did you score in Science in SEE?

    …….

22. Please categorize the educational background of your parents.

	SEE	Highschool Equiv...	Bachelors	Masters or above
Father	○	○	○	○
Mother	○	○	○	○

Markov Chain Monte Carlo Code:

```
library(haven)
library(tidyverse)
library(readxl)
#Importing and calculating parameters in Sample Data
priordist <- read_excel() #enter the file path
#Use log-normal distribution for monetary variables
priordist <- priordist %>%
  mutate(log_income <- log(income),
  log_scholarship <- log(scholarship + 1e-6),
  log_eduexp <- log(eduexp))
summary(priordist)
variances <- sapply(priordist, function(x) var(as.numeric(x), na.rm = TRUE))
print(variances)
#MCMC Model
library(rstan)
stan_data1 <- list(
  income = priordist$income,
  eduexp = priordist$eduexp,
  scholarship = priordist$scholarship,
  distance = priordist$distance,
  gpa = priordist$gpa,
  mathematics_see = priordist$mathematics_see,
  science_see = priordist$science_see,
  sex = priordist$sex,
  ethnicity = priordist$ethnicity,
  college_type = priordist$college_type,
  university = priordist$university,
  p_address = priordist$p_address,
  province = priordist$province,
  fgen_college = priordist$fgen_college,
  degree = priordist$degree,
  parental_expectations = priordist$parental_expectations,
  peer_influence = priordist$peer_influence,
```

```
  intrinsic_motiv = priordist$intrinsic_motiv,
  social_reputation = priordist$social_reputation,
  priorstem_member = priordist$priorstem_member,
  house_type = priordist$house_type,
  relocation = priordist$relocation,
  government_funding = priordist$government_funding,
  stem_activity_school = priordist$stem_activity_school,
  father_edu = priordist$father_edu,
  mother_edu = priordist$mother_edu
)
mcmc_model2 <- "
data {
   real  income[55];
   real  eduexp[55];
   real  scholarship[55];
   real<lower=0>  distance[55];
   real<lower=1.6, upper=4> gpa[55];
   real<lower=1.6, upper=4> mathematics_see[55];
   real<lower=1.6, upper=4> science_see[55];
   int<lower=1, upper=2> sex[55];
   int<lower=1, upper=4> ethnicity[55];
   int<lower=1, upper=2> college_type[55];
   int<lower=1, upper=5> university[55];
   int<lower=1, upper=4> p_address[55];
   int<lower=1, upper=7> province[55];
   int<lower=1, upper=2> fgen_college[55];
   int<lower=1, upper=4> degree[55];
   int<lower=1, upper=5> parental_expectations[55];
   int<lower=1, upper=5> peer_influence[55];
   int<lower=1, upper=5> intrinsic_motiv[55];
   int<lower=1, upper=5> social_reputation[55];
   int<lower=1, upper=2> priorstem_member[55];
   int<lower=1, upper=3> house_type[55];
   int<lower=1, upper=2> relocation[55];
```

```
    int<lower=1, upper=2> government_funding[55];
    int<lower=1, upper=2> stem_activity_school[55];
    int<lower=1, upper=4> father_edu[55];
    int<lower=1, upper=4> mother_edu[55];
}
parameters{
    real mu_income;
    real sigma_income;
    real mu_eduexp;
    real sigma_eduexp;
    real mu_scholarship;
    real<lower=0>  sigma_scholarship;
    real<lower=0> mu_distance;
    real<lower=0> sigma_distance;
    real<lower=0>  mu_gpa;
    real<lower=0> sigma_gpa;
    real<lower=0>  mu_mathematics_see;
    real<lower=0> sigma_mathematics_see;
    real<lower=0>  mu_science_see;
    real<lower=0> sigma_science_see;
    simplex[2] theta_sex;
    simplex[4] theta_ethnicity;
    simplex[2] theta_college_type;
    simplex[5] theta_university;
    simplex[4] theta_p_address;
    simplex[7] theta_province;
    simplex[2] theta_fgen_college;
    simplex[4] theta_degree;
    simplex[5] theta_parental_expectations;
    simplex[5] theta_intrinsic_motiv;
    simplex[5] theta_peer_influence;
    simplex[5] theta_social_reputation;
    simplex[2] theta_priorstem_member;
    simplex[3] theta_house_type;
```

```
    simplex[2] theta_relocation;
    simplex[2] theta_government_funding;
    simplex[2] theta_stem_activity_school;
    simplex[4] theta_father_edu;
    simplex[4] theta_mother_edu;
}
model{
    mu_income ~ normal(116655, 50000);
    sigma_income ~ cauchy(0,1);
    income ~ normal(mu_income, sigma_income);

    mu_eduexp ~ normal(217809, 200000);
    sigma_eduexp ~ cauchy(0,1);
    eduexp ~ normal(mu_eduexp, sigma_eduexp);

    mu_scholarship ~ normal(149702, 1810000);
    sigma_scholarship ~ cauchy(0,1);
    scholarship ~ normal(mu_scholarship, sigma_scholarship);



    mu_distance ~ normal(63.016, 31010.9);
    sigma_distance ~ cauchy(0,1);
    distance ~ normal (mu_distance, sigma_distance);

    mu_gpa ~ normal(3.41, 0.105);
    sigma_gpa ~ cauchy(0,1);
    gpa ~ normal(mu_gpa, sigma_gpa);

    mu_mathematics_see ~ normal(3.831, 0.088);
    sigma_mathematics_see ~ cauchy(0,1);
    mathematics_see ~ normal(mu_mathematics_see, sigma_mathematics_see);

    mu_science_see ~ normal(3.754, 0.0963);
    sigma_science_see ~ cauchy(0,1);
```

```
    science_see ~ normal(mu_science_see, sigma_science_see);

    theta_sex ~ dirichlet([0.42, 0.58]);
    theta_ethnicity ~ dirichlet([0.60, 0.22, 0.14, 0.04]);
    theta_college_type ~ dirichlet([0.36, 0.64]);
    theta_university ~ dirichlet([0.47, 0.42, 0.04, 0.02, 0.05]);
    theta_p_address ~ dirichlet([0.33, 0.07, 0.51, 0.09]);
    theta_province ~ dirichlet([0.16, 0.02, 0.58, 0.05, 0.13, 0.02, 0.4]);
    theta_fgen_college ~ dirichlet([0.44, 0.56]);
    theta_degree ~ dirichlet([0.24, 0.14, 0.33, 0.29]);
    theta_parental_expectations ~ dirichlet([0.2, 0.22, 0.31, 0.18, 0.09]);
    theta_intrinsic_motiv ~ dirichlet([0.44, 0.2, 0.16, 0.13, 0.07]);
    theta_peer_influence ~ dirichlet([0.09, 0.07, 0.25, 0.35, 0.24]);
    theta_social_reputation ~ dirichlet([0.22, 0.09, 0.22, 0.29, 0.18]);
    theta_priorstem_member ~ dirichlet([0.42, 0.58]);
    theta_house_type ~ dirichlet([0.82, 0.16, 0.02]);
    theta_relocation ~ dirichlet([0.42, 0.58]);
    theta_government_funding ~ dirichlet([0.2, 0.8]);
    theta_stem_activity_school ~ dirichlet([0.47, 0.53]);
    theta_father_edu ~ dirichlet([0.24, 0.25, 0.29, 0.22]);
    theta_mother_edu ~ dirichlet([0.36, 0.29, 0.22, 0.13]);
}

generated quantities {
    real income_generated[1];
    real eduexp_generated[1];
    real scholarship_generated[1];
    real gpa_generated[1];
    real mathematics_see_generated[1];
    real science_see_generated[1];
    real distance_generated[1];
    int sex_generated[1];
    int ethnicity_generated[1];
    int college_type_generated[1];
```

```
    int university_generated[1];
    int p_address_generated[1];
    int province_generated[1];
    int fgen_college_generated[1];
    int degree_generated[1];
    int parental_expectations_generated[1];
    int peer_influence_generated[1];
    int intrinsic_motiv_generated[1];
    int social_reputation_generated[1];
    int priorstem_member_generated[1];
    int house_type_generated[1];
    int relocation_generated[1];
    int government_funding_generated[1];
    int stem_activity_school_generated[1];
    int father_edu_generated[1];
    int mother_edu_generated[1];
    for (i in 1:1) {
        income_generated[i] = normal_rng(mu_income, sigma_income);
        eduexp_generated[i] = normal_rng(mu_eduexp, sigma_eduexp);
        scholarship_generated[i] = normal_rng(mu_scholarship, sigma_scholarship);
        gpa_generated[i] = normal_rng(mu_gpa, sigma_gpa);
        mathematics_see_generated[i] = normal_rng(mu_mathematics_see,
sigma_mathematics_see);
        science_see_generated[i] = normal_rng(mu_science_see, sigma_science_see);
        distance_generated[i] = normal_rng(mu_distance, sigma_distance);
        sex_generated[i] = categorical_rng(theta_sex);
        ethnicity_generated[i] = categorical_rng(theta_ethnicity);
        college_type_generated[i] = categorical_rng(theta_college_type);
        university_generated[i] = categorical_rng(theta_university);
        p_address_generated[i] = categorical_rng(theta_p_address);
        province_generated[i] = categorical_rng(theta_province);;
        fgen_college_generated[i] = categorical_rng(theta_fgen_college);
        degree_generated[i] = categorical_rng(theta_degree);
        parental_expectations_generated[i] = categorical_rng(theta_parental_expectations);
```

```
    peer_influence_generated[i] = categorical_rng(theta_peer_influence);
    intrinsic_motiv_generated[i] = categorical_rng(theta_intrinsic_motiv);
    social_reputation_generated[i] = categorical_rng(theta_social_reputation);
    priorstem_member_generated[i] = categorical_rng(theta_priorstem_member);
    house_type_generated[i] = categorical_rng(theta_house_type);
    relocation_generated[i] = categorical_rng(theta_relocation);
    government_funding_generated[i] = categorical_rng(theta_government_funding);
    stem_activity_school_generated[i] = categorical_rng(theta_stem_activity_school);
    father_edu_generated[i] = categorical_rng(theta_father_edu);
    mother_edu_generated[i] = categorical_rng(theta_mother_edu);
}
}  "


fit <- stan(model_code = mcmc_model2, data = stan_data1, iter = 5000, chains = 4)

generated_quantities <- extract(fit, pars = c(
  "income_generated", "eduexp_generated", "scholarship_generated",
  "sex_generated", "ethnicity_generated", "college_type_generated",
  "university_generated", "p_address_generated", "province_generated",
  "fgen_college_generated", "degree_generated", "parental_expectations_generated",
  "peer_influence_generated", "intrinsic_motiv_generated", "social_reputation_generated",
  "priorstem_member_generated", "house_type_generated", "relocation_generated",
  "government_funding_generated", "stem_activity_school_generated",
  "father_edu_generated", "mother_edu_generated", "gpa_generated",
"mathematics_see_generated", "science_see_generated", "distance_generated"
))

postdist <- data.frame(
  income = generated_quantities$income_generated,
  eduexp = generated_quantities$eduexp_generated,
  scholarship = generated_quantities$scholarship_generated,
  gpa = generated_quantities$gpa_generated,
  mathematics_see = generated_quantities$mathematics_see_generated,
```

```
  science_see = generated_quantities$science_see_generated,
  distance = generated_quantities$distance_generated,
  sex = generated_quantities$sex_generated,
  ethnicity = generated_quantities$ethnicity_generated,
  college_type = generated_quantities$college_type_generated,
  university = generated_quantities$university_generated,
  p_address = generated_quantities$p_address_generated,
  province = generated_quantities$province_generated,
  fgen_college = generated_quantities$fgen_college_generated,
  degree = generated_quantities$degree_generated,
  parental_expectations = generated_quantities$parental_expectations_generated,
  peer_influence = generated_quantities$peer_influence_generated,
  intrinsic_motiv = generated_quantities$intrinsic_motiv_generated,
  social_reputation = generated_quantities$social_reputation_generated,
  priorstem_member = generated_quantities$priorstem_member_generated,
  house_type = generated_quantities$house_type_generated,
  relocation = generated_quantities$relocation_generated,
  government_funding = generated_quantities$government_funding_generated,
  stem_activity_school = generated_quantities$stem_activity_school_generated,
  father_edu = generated_quantities$father_edu_generated,
  mother_edu = generated_quantities$mother_edu_generated
)

postdist <- postdist %>%
  mutate(income = ifelse(income < 0, income * (-1), income),
         eduexp = ifelse(eduexp < 0, eduexp * (-1), eduexp),
         scholarship = ifelse(scholarship < 0, 0, scholarship),
         distance = ifelse(distance < 0, distance * (-1), distance)
  )
postdist <- postdist[postdist$gpa <= 4, ]
postdist <- postdist[postdist$science_see <= 4, ]
postdist <- postdist[postdist$mathematics_see <= 4, ]

finaldf <- rbind(postdist, postdist1)
```

```
save(finaldf, file = "final.DTA")

#Dummy variables
finaldf <- finaldf %>%
  mutate(
    sex = ifelse(sex == 2, 0, sex),
    fgen_college = ifelse(fgen_college == 2, 0, fgen_college),
    priorstem_member = ifelse(priorstem_member == 2, 0, priorstem_member),
    relocation = ifelse(relocation == 2, 0, relocation),
    government_funding = ifelse(government_funding == 2, 0, government_funding),
    stem_activity_school = ifelse(stem_activity_school == 2, 0, stem_activity_school)
  )

#Multinomial Logistic Regression
library(nnet)
library(broom)
library(writexl)
#setting degree = 2 as reference level
finaldf$degree <- as.factor(finaldf$degree)
finaldf$degree <- relevel(finaldf$degree, ref = "1")
levels(finaldf$degree)

regmodel <- multinom ( degree ~ income + eduexp + scholarship + gpa +
                       mathematics_see + science_see + distance + sex +
                      + ethnicity + fgen_college + parental_expectations +
                       peer_influence + intrinsic_motiv + social_reputation +
                       priorstem_member + relocation + government_funding +
                       stem_activity_school + father_edu + mother_edu
                     , data = finaldf
                       )

summary(regmodel)
#Exporting the regression result
results <- tidy(regmodel)
```

```
write_xlsx(results, "regressionresultfinal.xlsx")

finaldf <- finaldf %>%
mutate(income = income/100,
            eduecp = eduexp/100,
             scholarship = scholarship/100
            )

#Autocorrelation
mcmc_samples <- as.mcmc(finaldf)
pdf("autocorrelation_plots.pdf")
mcmc_acf(mcmc_samples)
dev.off()

#Heidelberger and Welch Diagnostic
pdf("heidel_diag.pdf")
heidel_diag <- heidel.diag(mcmc_samples)
dev.off()

#ANOVA
library(stats)
income_list <- split(finaldf$income, finaldf$degree)
gpa_list <- split(finaldf$gpa, finaldf$gpa)

income_anova <- aov(income~degree, data=finaldf)
gpa_anova <- aov(gpa ~ degree, data=finaldf)
summary(income_anova)
summary(gpa_anova)

kruskal.test(income ~ degree, data = finaldf)
kruskal.test(gpa ~ degree, data = finaldf)
```

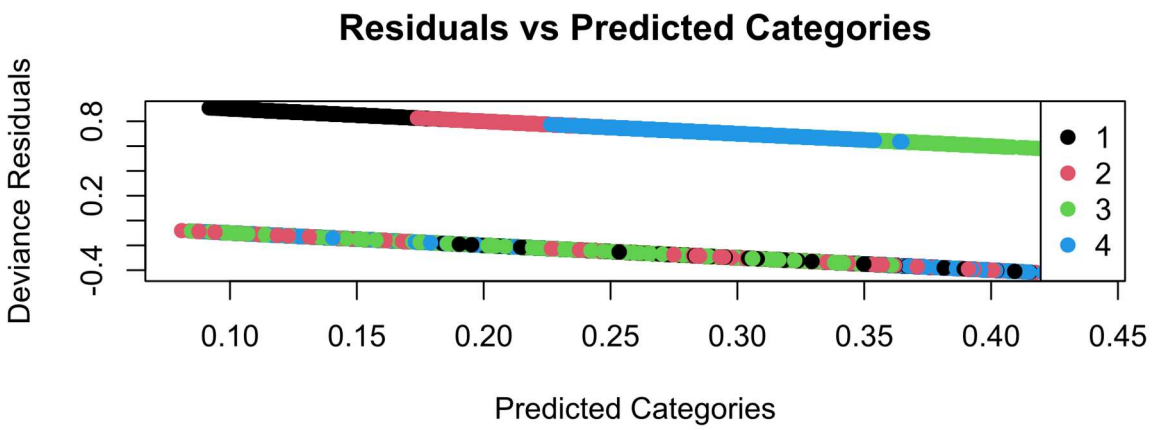
Heidelberger and Welch Diagnostic

	Stationarity test	start iteration	p value
income	passed	1	0.9677
eduexp	passed	1	0.9159
scholarship	passed	1	0.1372
gpa	passed	1	0.2700
mathematics_see	passed	1	0.1064
science_see	passed	1	0.3740
distance	passed	1	0.0969
sex	passed	1	0.7593
ethnicity	passed	1	0.8213
college_type	passed	1	0.1634
university	passed	1	0.2776
p_address	passed	1	0.2790
province	passed	1	0.6106
fgen_college	passed	1	0.6978
degree	passed	1	0.4365
parental_expectations	passed	1	0.2549
peer_influence	passed	1	0.8011
intrinsic_motiv	passed	1	0.7109
social_reputation	passed	1	0.0994
priorstem_member	passed	1	0.7711
house_type	passed	1	0.1204
relocation	passed	1	0.8085
government_funding	passed	1	0.8387
stem_activity_school	passed	1	0.8559
father_edu	passed	1	0.9994
mother_edu	passed	1	0.9394
eduecp	passed	1	0.9159

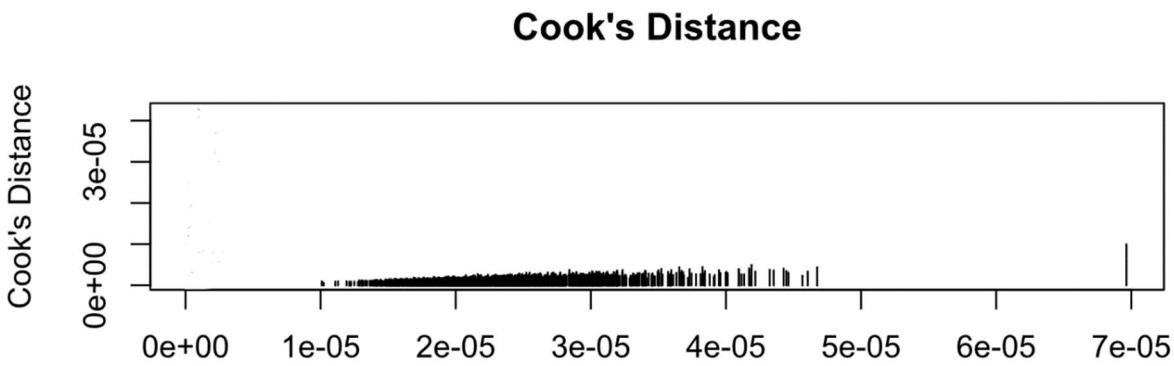
	Halfwidth test	Mean	Halfwidth
income	passed	1.38e+00	1.81e-02
eduexp	passed	2.72e+05	3.63e+03
scholarship	passed	2.56e+00	5.75e-02
gpa	passed	3.38e+00	5.44e-03
mathematics_see	passed	3.68e+00	4.14e-03
science_see	passed	3.63e+00	4.49e-03
distance	failed	1.06e+02	1.20e+01
sex	passed	4.11e-01	9.38e-03
ethnicity	passed	1.63e+00	1.68e-02
college_type	passed	1.63e+00	9.21e-03
university	passed	1.78e+00	1.93e-02
p_address	passed	2.36e+00	1.98e-02
province	passed	4.20e+00	3.95e-02
fgen_college	passed	4.50e-01	9.11e-03

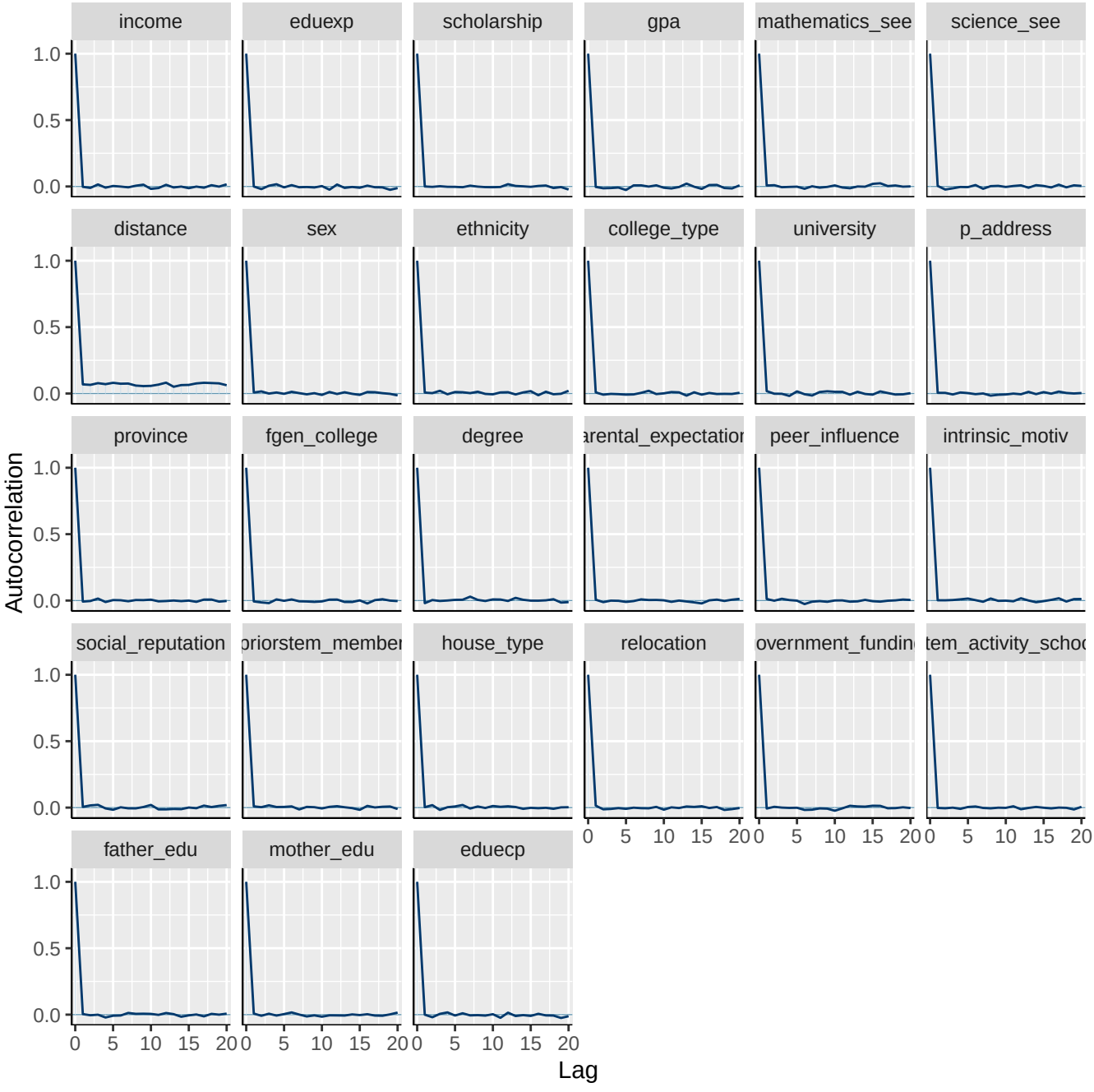
```
degree                 passed    2.77e+00 1.88e-02
parental_expectations  passed    2.74e+00 2.34e-02
peer_influence         passed    3.59e+00 2.26e-02
intrinsic_motiv        passed    2.20e+00 2.49e-02
social_reputation      passed    3.13e+00 2.81e-02
priorstem_member       passed    4.21e-01 9.41e-03
house_type             passed    1.21e+00 9.47e-03
relocation             passed    4.19e-01 9.54e-03
government_funding     passed    2.02e-01 7.65e-03
stem_activity_school   passed    4.84e-01 9.52e-03
father_edu             passed    2.50e+00 2.07e-02
mother_edu             passed    2.12e+00 1.97e-02
eduecp                 passed    2.72e+03 3.63e+01
```

Residuals vs Predicted Categories Plot



Cooks Distance




income
eduexp
scholarship
gpa
mathematics_see
science_see
distance
sex
ethnicity
college_type
university
p_address
province
fgen_college
degree
arental_expectatior
peer_influence
intrinsic_motiv
social_reputation
priorstem_member
house_type
relocation
overnment_fundin
tem_activity_schoo
father_edu
mother_edu
eduecp
Autocorrelation
Lag
1.0
0.5
0.0
0
5
10
15
20